

Math Review

ECON 97: Economic Toolkit

Patrick Molligo

UCLA

Spring 2026

Agenda

- ① Course overview
- ② Math review



<https://join.iclicker.com/RDNG>

Have you taken probability/stats before?

- A. Yes, college level
- B. Yes, high school level
- C. Never

What is Econ 97?

Main Goal: support you for success in Econ 41

- Review concepts
- Help with homework
- Exam prep

Additionally:

- Practice coding/data analysis
- Guest speakers

Diagnostic check: factorials

Compute the following:

$$5! =$$

$$0! =$$

$$\frac{9!}{6!} =$$

Diagnostic check: combinations

Compute the following (note: $\binom{n}{k} = {}_nC_k$):

$$\binom{5}{2} =$$

$$\binom{7}{7} =$$

$$\binom{9}{1} =$$

Diagnostic check: sets

Let $A = \{1, 3, 5, 7\}$, $B = \{0, 2, 5, 6, 7\}$, and $C = \{10, 11\}$.

Compute the following:

$$A \cap B =$$

$$A \cup B =$$

$$A \cup C =$$

Diagnostic check: sums

Compute the following:

$$\sum_{x=0}^5 x =$$

$$\sum_{x=0}^5 x^2 =$$

$$\sum_{x=0}^5 5 =$$

$$\sum_{x=0}^5 5x + 1 =$$

Diagnostic check: integrals

Compute the following:

$$\int_0^1 x dx =$$

$$\int_0^1 5 dx =$$

$$\int_0^1 x^2 dx =$$

Diagnostic check: probability

Flip a coin 3 times. What is the probability of 3 heads?

What is the probability of drawing a King or a Jack from a deck of cards?

Roll a 6-sided die 6 times. What is the probability of getting all different numbers?

Diagnostic check: statistics

Compute the sample mean, variance, and standard deviation of the following numbers:

$$\{1, 2, 3, 4, 5\}$$